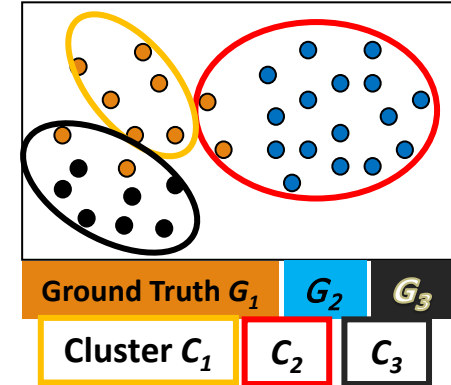


Information Theory-Based Methods (I)

Conditional Entropy

- A clustering can be regarded as a compressed representation of a given set of objects
- The better the clustering results approach the ground-truth, the less amount of information is needed
- This idea leads to the use of *conditional entropy*



Information Theory-Based Methods (I)

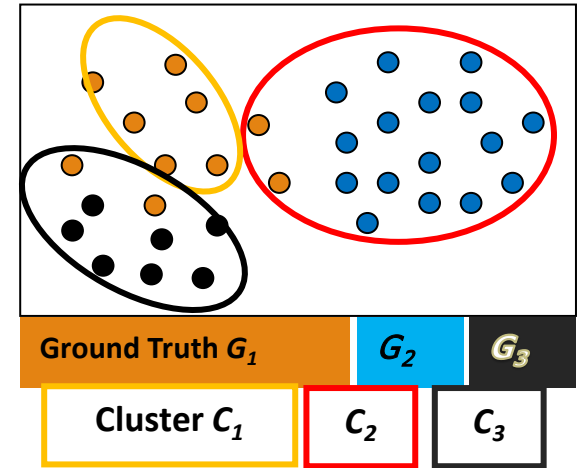
Conditional Entropy

- Entropy of clustering C: $H(C) = -\sum_{i=1}^m \frac{|C_i|}{n} \log \frac{|C_i|}{n}$
- Entropy of ground truth G: $H(G) = -\sum_{i=1}^l \frac{|G_i|}{n} \log \frac{|G_i|}{n}$
- Conditional entropy of G given cluster C_i :

$$H(G|C_i) = -\sum_{j=1}^l \frac{|C_i \cap G_j|}{|C_i|} \log \frac{|C_i \cap G_j|}{|C_i|}$$

- Conditional entropy of G given clustering C:

$$H(G|C) = \sum_{i=1}^m \frac{|C_i|}{n} H(G|C_i) = -\sum_{i=1}^m \sum_{j=1}^l \frac{|C_i \cap G_j|}{n} \log \frac{|C_i \cap G_j|}{|C_i|}$$



Example

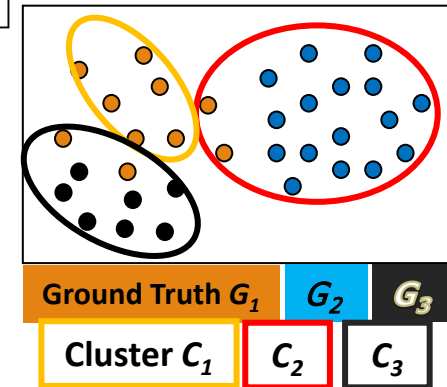
- Consider 11 objects

Object	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>	<i>k</i>
Ground truth $\mathcal{G}$	G_1	G_1	G_1	G_1	G_1	G_1	G_2	G_2	G_2	G_2	G_3
Clustering $\mathcal{C}_1$	C_1	C_1	C_1	C_1	C_2	C_2	C_3	C_3	C_3	C_3	C_4
Clustering $\mathcal{C}_2$	C_1	C_1	C_2	C_2	C_2	C_3	C_1	C_2	C_2	C_1	C_3

Purity for clustering $\mathcal{C}_1 = 1/11$ ($4 + 2 + 4 + 1$) = $11/11 = 1$;
Purity for clustering $\mathcal{C}_2 = 1/11$ ($2 + 3 + 1$) = $6/11$

$$H(\mathcal{G}|\mathcal{C}_1) = -\left(\frac{4}{11} \log \frac{4}{4} + \frac{2}{11} \log \frac{2}{2} + \frac{4}{11} \log \frac{4}{4} + \frac{1}{11} \log \frac{1}{1}\right) = 0$$

$$H(\mathcal{G}|\mathcal{C}_2) = -\left(\frac{2}{11} \log \frac{2}{4} + \frac{2}{11} \log \frac{2}{4} + \frac{3}{11} \log \frac{3}{5} + \frac{2}{11} \log \frac{2}{5} + \frac{1}{11} \log \frac{1}{2} + \frac{1}{11} \log \frac{1}{2}\right) \\ = 0.297$$



Note: conditional entropy cannot detect the issue that $\mathcal{C}_1$ splits the objects in G into two clusters